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## Fwd: Some further questions on PISA results and their interpretation

1 message

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**Aryan T** <athanki5670@gmail.com> 1 February 2024 at 14:47
To: "tompiccolo50@gmail.com" <tompiccolo50@gmail.com>

### Forwarded Conversation

**Subject: Some further questions on PISA results and their interpretation**
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From: **Aryan T** <athanki5670@gmail.com>
Date: Sun, 20 Aug 2023 at 10:53
To: <J.Jerrim@ioe.ac.uk>

Hello

I hope you are well

I had some more questions on PISA results:

The England 2006 PISA data did not suffer from bias like the previous years, so is the average score by grade for PISA 2006 (627= A*… etc) a true representation of the ability of different GCSE grade students? The score for an A* in other years where there was bias was lower (608)

For Korea, the 75th fpercentile was 606, 612, 609 for 2003, 2006, 2009 respectively but then jumped to 624 in 2012, and then fell again back to pre 2012 levels. Why was this?

From the PISA 2018 datasets I saw that Korea had a participation rate of 100%, however it has been stated in your research on FFTL Data Labs as 86%. Why is this?

What were the actual participation rates for Korea in PISA 2012, 2009, and 2006?

Canada had a 86% participation rate in PISA 2018 according to the “coverage index 3: coverage rate of 15 year old population”

In your research it states that it is only 63%, however. Why is this?

For Belgium, in 2003 the 75th percentile was 611. In 2006 it was significantly lower- despite the participation rate according to “coverage index 3: coverage rate of 15 year old population” being roughly the same.

Or is this wrong? Is coverage index/coverage rate measuring something else?

Also, for Switzerland over time:

Switzerland:

Maths 75th percentile:

2003: 595 (1.04)
2006: 600 (1.02)
2009: 604 (0.89)

Are the numbers in brackets above indicative of the participation rate decreasing?

Many thanks

Aryan Thanki

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From: **Jerrim, John** <j.jerrim@ucl.ac.uk>
Date: Sun, 20 Aug 2023 at 11:52
To: Aryan T <athanki5670@gmail.com>

My thoughts below:

The England 2006 PISA data did not suffer from bias like the previous years, so is the average score by grade for PISA 2006 (627= A*… etc) a true representation of the ability of different GCSE grade students? The score for an A* in other years where there was bias was lower (608)

JJ: How do you know there is no bias? They met the PISA standards. But that's a different thing. It's hard to claim no bias at all. Or even (much) less bias than previously.

For Korea, the 75th percentile was 606, 612, 609 for 2003, 2006, 2009 respectively but then jumped to 624 in 2012, and then fell again back to pre 2012 levels. Why was this?

JJ: don't know for sure. But the way Pisa scores are constructed means I am a bit more sceptical of reliability at top end of distribution. Probably a lot of noise going on here.

From the PISA 2018 datasets I saw that Korea had a participation rate of 100%, however it has been stated in your research on FFTL Data Labs as 86%. Why is this?

JJ: That would be it's response rate, probably after replacement schools are included. From memory, I reported pre-replacement response rates (which is actually more appropriate in my view). I also consider things like out of school population and exclusions.

What were the actual participation rates for Korea in PISA 2012, 2009, and 2006?

JJ: See the OECD technical reports. I think I give them in my paper(s) as well.

Canada had a 86% participation rate in PISA 2018 according to the “coverage index 3: coverage rate of 15 year old population”

In your research it states that it is only 63%, however. Why is this?

JJ: Different way of calculating. Including I use before replacement school response rates (and a couple of other things from memory).

For Belgium, in 2003 the 75th percentile was 611. In 2006 it was significantly lower- despite the participation rate according to “coverage index 3: coverage rate of 15 year old population” being roughly the same.

JJ: Could well be. Hard to know for sure without further data (like we have in England) to try and work out the potential bias at different parts of the distribution.

Or is this wrong? Is coverage index/coverage rate measuring something else?

JJ: It's trying to measure something different than the 75th percentile, that's for sure.

Also, for Switzerland over time:

Switzerland:

Maths 75th percentile:

2003: 595 (1.04)
2006: 600 (1.02)
2009: 604 (0.89)

Are the numbers in brackets above indicative of the participation rate decreasing?

JJ: I don't know what the numbers in brackets are.

Hope the above is some help.

John

**Check out my website! www.johnjerrim.com**

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**From:** Aryan T <athanki5670@gmail.com>
**Sent:** 20 August 2023 10:53
**To:** Jerrim, John <j.jerrim@ucl.ac.uk>
**Subject:** Some further questions on PISA results and their interpretation

⚠ Caution: External sender

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From: **Aryan T** <athanki5670@gmail.com>
Date: Sun, 20 Aug 2023 at 18:37
To: Jerrim, John <j.jerrim@ucl.ac.uk>

Hello

I hope you are well, and many thanks for your email.

I have many questions to ask, however there were some things I wanted to discuss with you.

What do you think the true ability of our top performers is? You must be aware of Attainment 8 scores, where say a score of 90 corresponds to the candidate gaining all 9’s (a 9 is above an A*) in Maths, English Language, Literature, the Sciences, Humanities and MFL subjects. Would it be fair to say even these top performers would not gain high PISA scores in Mathematics, say?

Would they not be able to compete with say, the top performers from South Korea or other Asian countries?

I ask this as I have investigated the average Attainment 8 score for candidates in selective or independent schools.

I investigated this for the academic year 2018-19- meaning these students would have taken part in PISA 2018.

Their average Attainment 8 score was in the high 70’s- around 77-78. This implies that in the core subjects- e.g Mathematics, candidates in selective or independent schools who took part in PISA 2018 gained grades 7 or 8 (A-A*) in Maths along with their other core academic subjects.

However their mean PISA Maths score for 2018 (which measured what grade they would go on to attain in Maths) was 504.

For an attainment 8 score of say, 78, as was their average, the PISA score would probably be between 561-585 for Mathematics.

Clearly there was some sort of negative bias occurring for our top performers, with the average scores not reflective of their actual performance.

I am asking this as I have put forward a FOI request to gov.uk asking for average PISA Maths scores for Attainment Eight scores of 80, 85 and 90.

What numbers should I be preparing to get in response? I myself attained 9’s in Maths, both English Language and Literature, all 3 sciences, geography, and a 7 in French and Art. Am I, and those like me, no match for candidates in certain countries?

I also wanted to ask about the state of exams in the UK. If I am not mistaken, the UK is one of the only countries in the world which allows calculator exams? Is this true?

If so, does this make our top performers inferior to those of other countries?

Also, to clarify some points you raised:

For Belgium, the 75th percentile dropped despite the overall coverage of 15 year olds being the same. If for several PISA cycles the 75th percentile for a country are roughly comparable and then in one, the score increases dramatically- is it comparable or subject to bias?

Also, in the previous email:

For Korea, the 75th percentile was 606, 612, 609 for 2003, 2006, 2009 respectively but then jumped to 624 in 2012, and then fell again back to pre 2012 levels. Why was this?

**JJ: don't know for sure. But the way Pisa scores are constructed means I am a bit more sceptical of reliability at top end of distribution. Probably a lot of noise going on here.**

Could you please expand on what you meant in your answer? (in bold for your reference)

Many thanks and I hope to receive a response

Aryan Thanki